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Ex. $x^2 - 13y^2 = 1$.

$$\sqrt{13} = 3 + (\sqrt{13} - 3)$$

$$\frac{1}{\sqrt{13}-3} = \frac{\sqrt{13}+3}{(\sqrt{13}-3)(\sqrt{13}+3)} = \frac{\sqrt{13}+3}{4} = 1 + \frac{\sqrt{13}-1}{4}$$

$$\frac{4}{\sqrt{13}-1} = \frac{\sqrt{13}+1}{3} = 1 + \frac{\sqrt{13}-2}{3}$$

$$\frac{3}{\sqrt{13}-2} = \frac{\sqrt{13}+2}{3} = 1 + \frac{\sqrt{13}-1}{3}$$

$$\frac{3}{\sqrt{13}-1} = \frac{\sqrt{13}+1}{4} = 1 + \frac{\sqrt{13}-3}{4}$$

$$\frac{4}{\sqrt{13}-3} = \sqrt{13}+3 = 6 + (\sqrt{13}-3)$$

So $\sqrt{13} = [3; \overline{1, 1, 1, 1, 6}]$.

	3	1	1	1	1	6	1	1	1	1
x	0	1	3	4	7	11	18	119	137	256	393	649
y	1	0	1	1	2	3	5	33	38	71	109	180
x^2-13y^2	-4	3	-3	4	-1	4	-3	3	-4	1		

So the fundamental solution to $x^2-13y^2=1$ is

$$(x,y)=(649,180).$$

All the solutions are given by

$$\pm(649+180\sqrt{13})^n,\quad n\in\mathbb{Z}.$$

In addition, the fundamental solution to $x^2-13y^2=-1$

is $(x,y)=(18,5)$.

All the solutions to $x^2 - 13y^2 = (-1)^n$ are

$$\pm(18 + 5\sqrt{13})^n \quad , \quad n \in \mathbb{Z}.$$

In particular,

$$(18 + 5\sqrt{13})^2 = 649 + 180\sqrt{13}.$$

Primes

Q: How many prime numbers are there?

Finite v.s ∞. ✓

Thm (Euclid) There are infinite many primes.

Idea: Assume there are only finitely many primes.

Find $n > 1$ indivisible by these primes, a contradiction,

because every $n > 1$ is divisible by a prime.

Say the primes are $p_1, \cdots, p_k$. Then

$p_i \mid p_1 \cdots p_k \quad (1 \le i \le k)$. Try $n = p_1 \cdots p_k + 1 > 1$.

Then $p_i \nmid n$; otherwise $p_i \mid n - p_1 \cdots p_k = 1$.

Pf. Assume there are only finitely many primes: $p_1, \cdots, p_k$.

Then $n = p_1 \cdots p_k + 1 > 1$ is indivisible by $p_1, \cdots, p_k$. This is impossible, since every $n > 1$ is divisible by a prime.

A contradiction. So there are ∞ many primes. $\square$

Variations.

Thm. There are infinite many primes $\equiv 3 \pmod 4$.

Pf. Assume there are only finitely many primes:

$$p_1, \cdots, p_k \equiv 3 \pmod 4.$$

Then $n = 4p_1 \cdots p_k - 1 > 4 - 1 = 3 > 1$ is indivisible by $p_1, \cdots, p_k$.

But this is impossible, since every $n > 1$ is divisible by a prime q, and $n \equiv 3 \pmod 4$ so that it will have at least one $q \equiv 3 \pmod 4$ (The product of primes $\equiv 1 \pmod 4$ is a number $\equiv 1 \pmod 4$). And $q \neq p_1, \ldots, p_k$ A contradiction. $\square$